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Step 1: Import Libraries

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from google.colab import files
import os
```

Step 2: Load Image and Preprocess

```
uploaded = files.upload()

filename = list(uploaded.keys())[0]

image = cv2.imread(filename)

if image is None:
    raise FileNotFoundError(f"Unable to load {filename}")

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(6,6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

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```
Saving 20250410_172721.jpg to 20250410_172721.jpg
```



Step 3: Convert to Grayscale and Apply Gaussian Blur

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray,(5,5),0)

plt.figure(figsize=(10,4))
```

```
plt.subplot(1,2,1)
plt.imshow(gray,cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(blur,cmap="gray")
plt.title("Gaussian Blurred Image")
plt.axis("off")

plt.show()
```



Step 4: Apply Global Thresholding

```
_, global_thresh = cv2.threshold(gray,127,255,cv2.THRESH_BINARY)

plt.figure(figsize=(6,6))
plt.imshow(global_thresh,cmap="gray")
plt.title("Global Thresholding")
plt.axis("off")
plt.show()
```



Step 5: Apply Otsu's Thresholding

```
_, otsu = cv2.threshold(
    blur,
    0,
```

```
    255,
    cv2.THRESH_BINARY + cv2.THRESH_OTSU
)

plt.figure(figsize=(6,6))
plt.imshow(otsu,cmap="gray")
plt.title("Otsu's Thresholding")
plt.axis("off")
plt.show()
```



Step 6: Apply Adaptive Thresholding

```
adaptive = cv2.adaptiveThreshold(
    gray,
    255,
    cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
    cv2.THRESH_BINARY,
    11,
    2
)

plt.figure(figsize=(6,6))
plt.imshow(adaptive,cmap="gray")
plt.title("Adaptive Thresholding")
plt.axis("off")
plt.show()
```



Step 7: Perform Watershed Segmentation

```
ret, thresh = cv2.threshold(
    gray,
    0,
    255,
    cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU
)

kernel = np.ones((3,3),np.uint8)

opening = cv2.morphologyEx(
    thresh,
    cv2.MORPH_OPEN,
    kernel,
    iterations=2
)

sure_bg = cv2.dilate(opening,kernel,iterations=3)

dist_transform = cv2.distanceTransform(
    opening,
    cv2.DIST_L2,
    5
)

ret, sure_fg = cv2.threshold(
    dist_transform,
    0.7*dist_transform.max(),
    255,
    0
)

sure_fg = np.uint8(sure_fg)

unknown = cv2.subtract(sure_bg,sure_fg)

ret, markers = cv2.connectedComponents(sure_fg)

markers = markers + 1

markers[unknown==255] = 0

watershed_image = image.copy()

markers = cv2.watershed(watershed_image,markers)
```

```
result = image_rgb.copy()

result[markers==-1] = [255,0,0]

plt.figure(figsize=(6,6))
plt.imshow(result)
plt.title("Watershed Segmentation")
plt.axis("off")
plt.show()
```

